

Should Dad take Depakote right now?

RECOMMENDATION

Defer divalproex. The OpenEvidence consultation completed 5/19 reinforces the original 5/13 recommendation: divalproex augmentation has no published RCT evidence base for unipolar TRD, is not in any major guideline for this indication, and carries a documented drug-induced parkinsonism signal that is particularly dangerous in Dad's specific prodromal-synucleinopathy profile. The drug ranked at or near the bottom of OE's geriatric TRD augmentation hierarchy. If "do more than ketamine" pressure persists, [bupropion augmentation \(handout #21\)](#) is a substantially stronger evidence-based alternative — direct OPTIMUM evidence (28.2% remission), no parkinsonism risk, and a more favorable cardiac and interaction profile. Chilton's role in raising divalproex is ambiguous (recommended? or noted on chart?) — that should be clarified directly in writing before any move. If divalproex is nonetheless pursued despite the evidence, the absolute requirement is a formal monitoring protocol with explicit motor, cognitive, and ammonia/CBC/LFT surveillance, and discontinuation triggers known to all family members upfront so any new symptom is surfaced rather than minimized.

WHAT THE 5/19 OPENEVIDENCE RESPONSE (#11) ADDED TO THE PICTURE

- **No RCT evidence for divalproex augmentation in unipolar TRD.** The only dedicated study is a small open-label trial (n=14, ages 19–59, no control group). Schaff 1993 (n=63) retrospectively reclassified 12 of the "unipolar" patients as covert cyclers. **Divalproex is not mentioned in any major guideline or review for unipolar TRD augmentation in older adults.**
- **OPTIMUM hierarchy for geriatric TRD augmentation** ranks aripiprazole augmentation (28.9% remission), bupropion augmentation (28.2%), and lithium augmentation (18.9% step 2) — **divalproex is not in any of the comparative arms.**
- **Valproate-induced parkinsonism is well-documented:** systematic prevalence rates 1.4–75% depending on assessment; insidious onset (mean 75.7 months in one series, but shorter latencies reported); dose-independent (occurs at therapeutic and subtherapeutic levels); recovery often prolonged (mean 4.5 months, range days to 2 years) and sometimes incomplete. Critically, **valproate can unmask subclinical dopaminergic degeneration** — UK Biobank data (n=1,433 PD cases, 8,598 controls) found sodium valproate persistently associated with incident PD even after excluding prescriptions within 5 years of diagnosis.
- **In Dad's profile specifically:** his prodromal-synucleinopathy picture (age 76, 8.5y RBD, constipation, gait/balance findings) places him in the highest-risk Arnaldi 2021 phenoconversion stratum (HR 5.71 combined criteria, two of three already met clinically). Introducing a drug known to unmask subclinical dopaminergic degeneration carries qualitatively different risk in this profile than in a patient without prodromal markers. Diagnostic confounding with the pending U-M Movement Disorders evaluation is the second-order concern — any new motor or cognitive finding could not be cleanly attributed.
- **Possible ketamine attenuation:** the one small (n=36) study examining ketamine response on valproate vs lithium maintenance showed a smaller antidepressant effect size on valproate (d = 0.79 vs 2.27), though not statistically significant. Mechanistic concern: valproate's GABAergic enhancement could theoretically dampen the glutamatergic mechanisms ketamine uses.
- **Significant doxepin interaction:** Unterecker 2014 — valproate increases doxepin + N-doxepin serum concentrations 2.2× via UGT inhibition. Clinically relevant given Dad's current doxepin 6 mg HS.
- **OE bottom line:** "Defer divalproex. Complete the U-M Movement Disorders and Cognitive Disorders evaluations first. Proceed with the planned IV ketamine course on June 8. If additional augmentation is needed beyond ketamine, bupropion augmentation or duloxetine has the strongest evidence-to-risk ratio for this patient profile."

Initiating divalproex (Depakote) at the proposed 125 mg twice daily (250 mg/day) with planned titration is **not supported by the evidence** and carries substantive risk in Dad's specific profile. Recommend deferring until after the neurology workup, the ketamine restart response read, and consultation with Gabe and Julane.

WHY THIS MATTERS FOR DAD SPECIFICALLY

Depakote is being prescribed primarily for prominent anxiety, with the specific clinical pattern of **early-morning awakening around 4 AM with anxious arousal** as a target. Dad also has multiple documented EMR findings that converge to make valproate-induced parkinsonism (VIP) and brain-volume risk higher than in the general elderly population: formal REM sleep behavior disorder (G47.52) diagnosed since 2017; drug-induced parkinsonism from aripiprazole (≤ 2 mg) in 2025; self-discontinued brexpiprazole for balance issues in March 2026; balance impairment (R26.89) coded nine months before any recent medication changes; "somewhat robotic" gait on his PCP physical exam 4/14/2026; brain fog (R41.89) coded multiple times; just-diagnosed methylmalonic acidemia (E71.120) with active B12 supplementation. He is currently in a severe depressive episode (PHQ-9 = 24), four days off a completed lorazepam taper, with ketamine restart scheduled in five days and an active neurology referral specifically to "rule out Parkinson's." Both the parkinsonism-risk profile *and* the lack of evidence for Depakote on the specific symptom pattern being targeted argue against its use here.

THE EVIDENCE

What might support starting Depakote	What argues against starting Depakote
One small study (open-label, meaning no placebo comparison) in 14 adults of mixed ages with hard-to-treat depression showed some improvement when Depakote was added to their regular antidepressant (Ghabrash 2016). This is the only published study suggesting a benefit, and it does not apply to Dad's age group or his dose.	No high-quality trials support it. No randomized placebo-controlled trials (the gold standard of medical evidence) have tested Depakote for hard-to-treat depression or anxiety in older adults — at any dose.
Depakote is widely prescribed off-label (i.e., for uses the FDA has not approved) at low doses and is generally tolerated short-term in adults without specific neurological vulnerabilities. This is genuine but weak evidence: it suggests the drug isn't so toxic that practice has abandoned it at this exposure level. It does not address whether it provides benefit for hard-to-treat depression, and tolerability in the general population may not transfer to Dad's specific profile.	No evidence at the dose Dad is prescribed. Standard treatment doses for Depakote's approved uses are typically 750–2,500 mg/day. At 250 mg/day, the drug's level in the blood doesn't reach the range at which any benefit has been documented.
—	Not recommended by any major clinical guideline. The American Psychiatric Association, the UK's NICE guidelines, Canada's CANMAT, the Maudsley Hospital prescribing guide, the American geriatric psychiatry association (AAGP), and the U.S. Department of Veterans Affairs all omit Depakote as an add-on treatment for hard-to-treat depression.
—	The "low dose protects against side effects" idea doesn't hold up. Depakote can cause Parkinson's-like symptoms (tremor, stiffness, slowed movement) as a side effect — a syndrome called valproate-induced parkinsonism. A 2016 review of 116 published cases found that in 113 of them, the patient's blood level of the

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	drug was at or below the standard "safe" range when these symptoms appeared.
—	A landmark 2011 trial (Tariot et al.) gave older adults with early Alzheimer's roughly the same dose Dad is prescribed (about 250 mg/day). Over 12 months: their brains shrank faster on MRI, and they scored worse on a routine cognitive screening test (the Mini-Mental State Examination) than those given placebo.
—	A large 2023 UK study (Belete et al., 1,433 patients newly diagnosed with Parkinson's disease) found that people who had taken Depakote were more likely to receive that diagnosis later — and the more total exposure to the drug they'd had, the stronger the link.
—	Would muddy the neurology evaluation. Dad's June neurology appointment is specifically to assess possible Parkinson's-spectrum disease. Depakote's side effects (movement problems, cognitive changes) look identical to those very findings — so the neurologist couldn't tell whether anything new came from the drug or from an underlying condition.
—	Would interfere with Dad's B12 follow-up. Depakote disrupts how the body processes folate and vitamin B12 (which is measured indirectly through a blood marker called methylmalonic acid, or MMA). Dad has an MMA blood test scheduled for August 6, 2026 to check whether his B12 supplementation is working. Starting Depakote would make those results impossible to interpret cleanly.
—	Dad's profile suggests a brain that is already vulnerable to drugs like this. He has REM Sleep Behavior Disorder (a sleep condition in which people physically act out their dreams) formally diagnosed on his chart since 2017 — about 8.5 years. Combined with the prior medication reactions and the recent gait findings, this points to underlying nervous-system susceptibility.
—	No evidence Depakote helps the specific sleep pattern being targeted — and it may actually make sleep worse. There is no randomized trial, clinical guideline, or expert-consensus statement supporting Depakote for early-morning awakening (terminal insomnia) with anxiety in older adults. The published polysomnography data on valproate suggest it actually <i>worsens</i> sleep quality by increasing the lightest stage of sleep (Stage 1) — the opposite of what's needed to stay asleep through the night. Early-morning awakening with morning anxiety is a classic feature of major depression itself, and the evidence-supported approach is to treat the depression — which is exactly what the ketamine restart will do.
—	Ketamine has documented evidence of improving exactly this sleep pattern. A 2022 study of 323 TRD patients receiving ketamine found that improvement in early-morning waking was

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	a significant partial mediator of both the antidepressant and the anti-suicidal effects. A 2021 analysis showed that sleep improvement within 24 hours of the first ketamine infusion predicts the overall antidepressant response. A 2026 systematic review of 26 studies (n=1,694) confirms ketamine's favorable effects on sleep quality in TRD. So Dad's documented prior dramatic response to ketamine — restart in five days — is likely to address this specific pattern directly.
—	Some of the current 4 AM awakening is probably benzodiazepine rebound that will resolve on its own. Dad finished his lorazepam taper four days ago after about three weeks of near-daily use. The 1–4 day window after stopping a short-acting benzodiazepine like lorazepam is exactly when rebound insomnia and rebound anxiety typically peak — and these symptoms usually resolve over the following 1–2 weeks without intervention. Adding Depakote now risks treating a problem that's already on its way to resolving by itself.

Evidence strength against starting Depakote: ★★★★★

Convergence of guideline absence, no RCT support, established mechanistic risk in a profile with multiple matched vulnerabilities, and active diagnostic workup that would be confounded.

RECOMMENDED NEXT STEPS

1. **Hold Depakote at the current dose pending review** — do not titrate the second-week increase as planned.
2. **Family / Julane / Gabe consultation** before any further decision on mood-stabilizer addition.
3. **Watch for any new motor, gait, sedation, or cognitive change** at any dose; if observed, discontinue and inform neurology.
4. **Continue B12 1000 mcg daily as prescribed**, and protect the 8/6/2026 MMA recheck from confounders.
5. **Proceed with sublingual ketamine restart May 18 as planned**; assess response over 2–3 sessions before any additional medication change.
6. **If Dr. Rubin requires a mood-stabilizer trial**, request a brief written rationale specifying expected mechanism of benefit at 250 mg/day, planned target dose, and stopping criteria.

BOTTOM LINE *The case for divalproex 250 mg/day in Dad's profile is unsupported by guidelines or RCT evidence — including for the specific early-morning awakening pattern being targeted — and is meaningfully contraindicated by his neurology workup, RBD history, B12 axis, and prior drug-induced parkinsonism. The early-morning awakening pattern itself is most likely a combination of depressive terminal insomnia (which ketamine restart should address directly) and benzodiazepine-rebound insomnia (which is expected to resolve over the next 1–2 weeks). Defer until the ketamine restart has been assessed and the K Clinic psychiatrist, Gabe, and Julane have reviewed.*

Key Studies & Findings

Ranked by authority · Each citation summarized for how it applies to Dad's case

TIER 1 — MAJOR CLINICAL GUIDELINES

★★★★★

Steffens DC. Treatment-Resistant Depression in Older Adults.

NEJM 2024;390:630–639. Read [here](#).

KEY FINDING The current NEJM review of TRD in adults ≥ 65 lists aripiprazole, bupropion, lithium, triiodothyronine, nortriptyline, MAOIs, esketamine, ECT, and TMS as recommended next-step strategies. Valproate is not mentioned.

HOW IT APPLIES The most current authoritative summary of late-life TRD treatment does not endorse valproate at any dose.

Lam RW, Kennedy SH et al. CANMAT 2023 update on adult MDD.

Can J Psychiatry 2024 (Lam RW et al.); summarized in Coles & Wise, *Am Fam Physician* 2025;112:458–461.

KEY FINDING Recommends aripiprazole as first-line augmentation; olanzapine, quetiapine, and risperidone as second-line. Valproate is not listed at any recommendation tier for unipolar MDD.

HOW IT APPLIES The Canadian guideline most widely used in North America does not support valproate for this indication.

McQuaid JR, Buelt A et al. VA/DoD 2022 MDD Guideline Synopsis.

Ann Intern Med 2022;175:1440–1451. Read [here](#).

KEY FINDING Does not include valproate among recommended augmentation strategies for unipolar MDD.

HOW IT APPLIES The Veterans Affairs / Department of Defense guideline — widely used for evidence-based late-life care — excludes valproate.

Taylor RW et al. Pharmacological Augmentation in Unipolar Depression: A Guide to the Guidelines.

Int J Neuropsychopharmacol 2020;23:587–625. Read [here](#).

KEY FINDING Systematic review of 10 major depression guidelines. Lithium and atypical antipsychotics were recommended by all 10. Valproate was not universally recommended by any.

HOW IT APPLIES Across guideline frameworks worldwide, valproate has no consistent endorsement for the indication Dr. Rubin is proposing.

TIER 2 — MAJOR RCTS IN TARGET POPULATION**Lenze EJ et al. Antidepressant Augmentation versus Switch in Treatment-Resistant Geriatric Depression (OPTIMUM).**

NEJM 2023;388:1067–1079. Read [here](#).

KEY FINDING Tested aripiprazole augmentation, bupropion augmentation, switch to bupropion, lithium augmentation, and switch to nortriptyline in adults ≥ 60 with TRD. Valproate was not included.

HOW IT APPLIES The landmark geriatric TRD trial of the past five years did not include valproate among options worth testing.

Tariot PN et al. Chronic Divalproex Sodium to Attenuate Agitation and Clinical Progression of Alzheimer Disease.

Arch Gen Psychiatry 2011;68:853–861. Read [here](#).

KEY FINDING $n=313$ randomized; MRI substudy ($n=89$). Divalproex (mean modal ~ 250 mg/day) for 24 months produced accelerated hippocampal volume loss (10.9–12.4% vs 5.6–6.3%), accelerated whole-brain loss (3.5% vs 1.4%), and faster MMSE decline (-3.9 vs -2.0 , $p=0.04$).

HOW IT APPLIES The single highest-quality RCT of low-dose valproate in cognitively vulnerable elderly used the exact dose Dad has been prescribed (250 mg/day) and showed brain-volume harm without antidepressant benefit. Dad's profile places him in or beyond the vulnerability category of the Tariot population.

TIER 3 — META-ANALYSES & LARGE OBSERVATIONAL**Belete D, Jacobs BM et al. Association Between Antiepileptic Drugs and Incident Parkinson Disease.**

JAMA Neurology 2023;80:183–187. Read [here](#).

KEY FINDING UK Biobank nested case-control study of 1,433 incident PD cases. Sodium valproate robustly associated with incident PD diagnosis. Association persisted after excluding prescriptions within 5 years of diagnosis and adjusting for epilepsy; strengthened with greater cumulative exposure.

HOW IT APPLIES The largest population-level study to date. Cumulative-exposure-dependent association argues against "short-term use is safe."

Strawbridge R et al. TRD augmentation systematic review.

Br J Psychiatry 2019. Read [here](#).

KEY FINDING Lithium had 13 placebo-controlled RCTs in unipolar TRD; valproate had zero placebo-controlled RCTs in pure unipolar TRD.

HOW IT APPLIES The asymmetry is not "weaker evidence" but "established evidence vs. no evidence."

Nuñez NA et al. Augmentation Strategies for Treatment-Resistant Major Depression: Network Meta-Analysis.

J Affect Disord 2022;302:385–400. Read [here](#).

KEY FINDING Network meta-analysis of 65 RCTs (N=12,415). Identified significant response for aripiprazole, brexpiprazole, quetiapine, lithium, liothyronine, modafinil, olanzapine/fluoxetine, cariprazine, and lisdexamfetamine. Valproate was not among the 19 agents with adequate evidence to be studied.

HOW IT APPLIES When the literature is comprehensively pooled, valproate does not even reach the threshold to enter the comparison.

Rodrigues NB et al. Do Sleep Changes Mediate the Anti-Depressive and Anti-Suicidal Response of Intravenous Ketamine in Treatment-Resistant Depression?

J Sleep Res 2022;31:e13400. Read [here](#).

KEY FINDING Study of 323 TRD patients receiving IV ketamine. Improvement in early-morning waking was a significant partial mediator of both the antidepressant and the anti-suicidal effects of ketamine. Each one-point improvement in total sleep score was associated with about 3.3 times higher odds of achieving treatment response/remission.

HOW IT APPLIES Direct evidence that ketamine specifically helps with early-morning waking — the symptom pattern Dr. Rubin is targeting with Depakote. Dad's documented prior dramatic ketamine response means the May 18 restart is the evidence-supported intervention for this exact symptom.

Boesjes R, Oosterveld C et al. Effects of Ketamine on Sleep and Circadian Rhythmicity in Major Depressive Disorder and Bipolar Disorder: A Systematic Review.

J Affect Disord 2026;409:121915. Read [here](#).

KEY FINDING Systematic review of 26 studies (1,694 patients). Ketamine is associated with favorable effects on subjective sleep quality. Baseline sleep disturbances and early sleep improvements may predict overall antidepressant response to ketamine.

HOW IT APPLIES The most current pooled evidence that ketamine improves sleep specifically in TRD patients. Strengthens the case that Dad's sleep pattern is best addressed via the ketamine restart, not via Depakote.

TIER 4 — MECHANISTIC / RISK-SIDE TRIALS



Brugger F, Bhatia KP, Besag FMC. Valproate-Associated Parkinsonism: A Critical Review.

CNS Drugs 2016;30:527–540. Read [here](#).

KEY FINDING Systematic review of 116 published VIP cases. In 113 of 116, plasma valproate concentrations were within or below the reference range when parkinsonism occurred. VIP odds ~5× higher with valproate than with other antiepileptics.

HOW IT APPLIES Directly undermines the "low dose protects" rationale. At 250 mg/day, Dad would be in the range where the majority of VIP cases have been described.

Fleisher AS et al. Chronic Divalproex Sodium Use and Brain Atrophy in Alzheimer Disease.

Neurology 2011;77:1263–1271. Read [here](#).

KEY FINDING The MRI substudy from the Tariot trial. Confirmed accelerated hippocampal and whole-brain volume loss at mean ~250 mg/day in cognitively vulnerable elderly.

HOW IT APPLIES Parallel evidence at the same dose Dad would be on.

Linnebank M et al. Antiepileptic Drugs Interact With Folate and Vitamin B12 Serum Levels.

Ann Neurol 2011;69:352–359. Read [here](#).

KEY FINDING n=2,730 AED-treated patients. Valproate-treated patients had lower folate and elevated homocysteine; B12 paradoxically slightly higher but with disrupted methylation pathways.

HOW IT APPLIES Confounds the planned 8/6/2026 MMA recheck — valproate's effect on one-carbon metabolism would make it impossible to interpret whether MMA normalization reflected B12 supplementation success or was offset by valproate.

Legros B, Bazil CW. Effects of Antiepileptic Drugs on Sleep Architecture: A Pilot Study.

Sleep Med 2003;4:51–55. Read [here](#).

KEY FINDING Polysomnography study comparing antiepileptic drugs' effects on sleep architecture. Valproic acid disrupted sleep by significantly increasing Stage 1 (light, fragmented) sleep compared to controls (16.8% vs. 7.7%), suggesting valproate may worsen overall sleep quality.

HOW IT APPLIES Counters any suggestion that Depakote could help Dad sleep through the night. The available sleep-architecture data on valproate point in the opposite direction — toward more fragmented, lighter sleep, not less.

TIER 5 — SMALL OPEN-LABEL & CASE-LEVEL EVIDENCE



Ghabrash MF et al. Valproate Augmentation in a Subgroup of Patients With Treatment-Resistant Unipolar Depression.

World J Biol Psychiatry 2016;17:165–170. Read [here](#).

KEY FINDING Open-label, n=14, ages 19–59. Valproate 375–1000 mg/day augmenting antidepressants. MADRS improvement at 1, 4, and 7 months. No control group.

HOW IT APPLIES The highest-quality evidence specifically for valproate in unipolar TRD is a 14-patient open-label trial in a mixed-age population at doses 1.5–4× what Dad has been prescribed.

Athauda D, Batley R, Ellis C. Clinically Silent Idiopathic Parkinson's Disease Unmasked by Valproate Use.

Aging Clin Exp Res 2015;27:387–390. Read [here](#).

KEY FINDING Case report of valproate unmasking previously asymptomatic Parkinson disease in an elderly patient. Only partial reversal after drug cessation.

HOW IT APPLIES Mechanistic illustration of the substrate-unmasking concern — directly relevant given Dad's prior DIP, balance findings, RBD history, and "robotic" gait observation.

Aliyev NA, Aliyev ZN. Anxiolytic effect of valproic acid in chronically anxious patients.

Eur Psychiatry 2008;23:109–114. Read [here](#).

KEY FINDING Single RCT of sustained-release valproate in 80 men (all male) with pure GAD, without elderly representation. Positive response, never replicated. Bahji 2024 overview of reviews found valproate's anxiety signal across all available studies was non-significant (SMD –0.32, p=0.12).

HOW IT APPLIES The single positive anxiety RCT exists but in a young-to-middle-aged male cohort at higher doses than Dad would receive — and the meta-analytic signal does not hold up.